

CLAYTON GROTH

Software Engineer

(608) 498-7370 | claytoncgrth@gmail.com | github.com/claytongroth | claytongrothdev.com

PROFESSIONAL SUMMARY

I am a Software Engineer with deep expertise in scalable web development, geospatial systems, data pipelines, data analysis, cyber security, and technical communication. I have a proven ability to design and implement software solutions that support diverse research and industry initiatives. In my career, I have consistently combined technical skills in science, software, and GIS systems with good leadership and empathetic communication. I am the person who can pick the tech stack, explain why, build it from the ground up, and convey how it works in simple terms.

RELEVANT EXPERIENCE

cQuant

January 2021 – Present

Senior Software Engineer | Remote

- Lead technical design and implementation of software architecture for production systems
- Built a robust renewable energy valuation tool from the ground up with geospatial tech, and energy grid based analysis
- Design and deploy containerized applications (Docker, Kubernetes) with infrastructure automation (Ansible, Helm)
- Develop high-performance production-grade data processing pipelines in Python, Go, and more for handling multi-dimensional environmental, financial and geospatial data
- Write and hold responsibility for countless lines of production code
- Debug, monitor, and test critical production systems for stability and security
- Provide technical leadership and mentoring to engineering team on software architecture, security, and best practices
- Collaborate with web and quantitative development teams to build and improve new and legacy software in Python, R, JavaScript, Go, Bash/Shell, SQL, and more

Digital Locale

June 2019 – January 2021

Full Stack Developer | Remote

- Developed travel/mapping application integrating multiple geospatial data sources with interactive visualization using React and Mapbox GL

- Built REST APIs and spatial database models using Sequelize ORM and PostgreSQL for location-based research and analysis
- Implemented geospatial analysis features for route optimization and spatial pattern detection
- Created responsive web interfaces for exploring geographic data and environmental patterns

State Cartographer's Office (UW Madison)

September 2018 – April 2019

GIS Developer | Madison, WI

- Designed and developed web-based geospatial applications for visualizing spatial and tabular data using ReactJS, enabling researchers and state agencies to analyze datasets
- Built Python-based spatial analysis tools for processing large geospatial datasets, supporting research projects in environmental science and public policy
- Developed complex queries with PostGIS for analyzing large spatial datasets for Wisconsin state government environmental monitoring programs
- Created interactive cartographic visualizations and data dashboards that translated complex spatial research into accessible tools for non-technical eyes
- Collaborated with academic researchers to understand requirements and deliver custom geospatial analysis solutions
- Supported research workflows by building reusable GIS tools and automated spatial data processing pipelines

Center for Sustainability and the Global Environment (UW Madison) September 2015 – April 2016

Student Lab Tech | Madison, WI

- Assisted in research tasks (mostly scripting)
- Wrote and assisted with various data-pipeline and atmospheric modeling scripts in R, Fortran, Java, and Bash/Shell
- Completed a thesis on various atmospheric correlates of Ozone with EPA & NOAA data

EDUCATION

University of Wisconsin-Madison

M.S. Geospatial Programming | September 2017 – August 2019

- Specialized in geospatial software development, spatial algorithms, spatial analysis, and GIS systems
- Extensive coursework in both frontend and backend development for research-grade spatial systems
- Graduate-level training in spatial statistics, environmental modeling, and scientific computing

University of Wisconsin-Madison

B.S. Environmental Science (Double Major: Philosophy) | September 2011 – May 2016

- Focus: Computational analysis of climatological and air quality data using spatial modeling for the SAGE (Center for Sustainability and the Global Environment)
 - Research on spatial patterns in environmental datasets and climate change impacts
 - Training in environmental systems, earth science, and data-driven environmental analysis
 - Philosophy: grad-level coursework in logic, probability theory, and epistemology
 - Published in an undergraduate journal and presented a paper on wellbeing metrics
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TECHNICAL SKILLS

Software Engineering:

- Languages: Python (expert), JavaScript/Node.js (expert), SQL (expert), Go, C++, Shell
- Databases: PostgreSQL/PostGIS (expert), Oracle SQL, MongoDB, SQLite
- Backend Development: Node.js, Express, REST APIs, GraphQL, Flask, Django
- Frontend Development: React (expert), Redux, TypeScript, responsive design
- Unix/Linux: Expert command-line proficiency, system administration, shell scripting
- Testing & Quality: Test-driven development, automated testing, CI/CD pipelines
- Documentation: Technical writing, API documentation, research software best practices
- Cyber Security: CISSP training, SAST/DAST scanning experience, deep knowledge of software vulnerabilities, how to track and mitigate them
- Version Control: Git, GitHub, collaborative research workflows, reproducible research

Research Computing & Data Science:

- Python (6+ years): NumPy, Pandas, SciPy, scikit-learn, Matplotlib, Jupyter, TensorFlow
- Scientific Computing: Large-scale data processing, parallel computing, data processing optimization for terabyte+ sized datasets
- Data Analysis: Time series analysis, statistical modeling, some machine learning applications
- R: Statistical analysis, spatial statistics, data visualization, environmental modeling

Geospatial:

- GIS Software: PostGIS, QGIS, ArcGIS, GDAL/OGR, geospatial data formats
- Python Geospatial Stack: Fiona, Shapely, GeoPandas, Rasterio, OSMNX
- Spatial Analysis: Spatial statistics, network analysis, terrain analysis, geocoding
- Mapping & Visualization: Mapbox GL, Leaflet, CartoDB, OpenLayers, web mapping
- Data Formats: GeoJSON, Shapefiles, GeoTIFF, NetCDF, KML

Infrastructure & DevOps:

- Containerization: Docker, Docker-Compose, reproducible and secure containerized environments
- Orchestration: Kubernetes, scalable deployment for research computing

- Configuration Management: Ansible, KubeCron, Cron
- Cloud Computing: AWS, Azure, production deployment and scaling
- Research Infrastructure: Supporting academic research workflows, data pipelines

Understanding of Research Environments:

- Academic research workflows and data management requirements
 - Collaborative research software development practices
 - Supporting diverse research initiatives across disciplines
 - Translating research requirements into software solutions
 - Working with faculty, researchers, and students through effective and empathetic communication
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TECHNICAL LEADERSHIP & COLLABORATION

- Software Architecture: Design high-level system architecture for industry applications
 - Technical Mentoring: Guide junior developers and aspiring developers in software best practices and career choices
 - Idea to Tech Translation: Convert complex research and industry needs into implementable software designs
 - Cross-disciplinary Communication: Bridge technical and domain expertise with quantitative analysts, environmental professionals, and researchers
 - Code Review: Ensure quality, security, and maintainability of software
 - Documentation: Create comprehensive technical and user documentation
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PROFESSIONAL DEVELOPMENT

- CISSP Training: Cybersecurity best practices for research data protection
 - Continuous Learning: Active engagement with philosophy and constant curiosity
 - Turing Mentor: Active engagement mentoring aspiring software engineers
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References available upon request